

Technical Datasheet for TnH Sensor Prob (Metal)

The TnH (Temperature and Humidity) Sensor Module provides highly accurate environmental readings over a standard I2C interface. Designed for straightforward, local integration, this sensor is highly efficient, boasts low power consumption, and is exceptionally easy to deploy in custom hardware builds.



Hardware & Design

- **Rugged Metal Probe:** The internal temperature and humidity sensing elements are securely housed within a durable, protective metal casing. This robust physical design shields the sensor from physical damage and environmental wear, making it suitable for a wide variety of indoor or enclosed monitoring applications.
- **I2C Communication:** This module utilizes a simple, two-wire I2C communication protocol (SDA, SCL). This makes it the perfect choice for short-distance, localized data acquisition where direct connection to a master microcontroller (such as a Raspberry Pi, Arduino, or custom DAQ board) is required.
- **Wiring Interface:** The probe features a flexible, multi-core pigtail cable with exposed wire leads, allowing technicians to quickly terminate and solder the connections directly to their expansion boards or terminal blocks.

Pin	Name	Color	Description
1	VDD	Red	Power Supply (3.3V/ 5V)
2	SDA	Yellow	I2C line SDA
3	SCL	Green	I2C line SCL
4	GND	Black	Common Ground

Specifications

General –

Power

Input Supply – 3.3 & 5V DC

Storage Temperature

-20 - 70° C (-4 ~ 158°F)

Storage Humidity

5 ~ 95 % RH, non – Condensing

Certifications


	Sensor Name	Temperature Measuring Range	Humidity Measuring range	AQI/ GAS/ Pressure	Measuring Accuracy	Supply (VDD)
☐	AHT20	-20 ~ 85°C	0~100% RH.	N/A	0.3°C, 2.0 %RH	3.3V/ 5V
☐	AHT21	-40 ~ 120°C	0~100% RH.	N/A	0.3°C, 2.0 %RH	3.3V/ 5V
☐	SHT40	-40 ~ 125°C	0~100% RH.	N/A	0.2°C, 1.8 %RH	3.3V
☐	SHT45	-40 ~ 125°C	0~100% RH.	N/A	0.1°C, 1.0 %RH	3.3V
☐	BME280	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa	1.0°C, 3.0 %RH, 1.0 %hPa	3.3V
☐	BME680	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa & 0 – 500 IAQ	1.0°C, 3.0 %RH, 0.6 %hPa, 15 %IAQ	3.3 V
☐	BME688	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa & 0 – 500 IAQ	0.5°C, 3.0 %RH, 0.6 %hPa, 15 %IAQ	3.3 V
☐	BME690	-40 ~ 85°C	0~100% RH.	300 - 1100 hPa & 0 – 500 IAQ	0.5°C, 3.0 %RH, 0.5 %hPa., 15 %IAQ	3.3 V

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